

AI-Based Coding Assessment Platform

Problem Statement and Literature Survey

1. Problem Statement

Programming education and coding assessments increasingly rely on online judges and automated grading systems to evaluate large numbers of student submissions. Traditional systems are efficient at checking compilation, test cases, and output correctness, but they often provide limited personalized feedback and may not adequately assess code quality, programming understanding, or the learner's actual ability.

The growing availability of Large Language Models (LLMs) such as ChatGPT introduces an additional challenge: a technically correct code submission may not always represent a student's own programming knowledge. At the same time, instructors need assessment systems that are scalable, fair, efficient, and capable of giving useful feedback without increasing manual grading workload.

The three selected research papers address these challenges from complementary perspectives: AI-powered code review and personalized feedback, automated programming assessment and grading, and assessment designs intended to reduce the impact of AI-generated solutions. However, each approach focuses on only part of the overall assessment problem.

Therefore, the proposed project, an AI-Based Coding Assessment Platform, aims to provide an integrated environment for coding assessment that combines automated code execution and evaluation, AI-assisted feedback, assessment of code quality and programming understanding, and mechanisms to support fair assessment in the era of generative AI.

2. Literature Survey

1. A GPT-Based Code Review System With Accurate Feedback for Programming Education

Author(s): Dong-Kyu Lee and Inwhee Joe (2025)

Algorithm / Methodology: GPT-4o-based code review system using a Code Review Module (CRM), Code Correctness Check Module (CCM), traditional test-case validation, LLM-based code assessment, and a Review Necessity Chain (RNC) to filter unnecessary reviews.

Advantages: Immediate personalized feedback; combines test-case checking with AI-based code assessment; identifies issues beyond simple output mismatches; reduces unnecessary LLM usage; supports learner-friendly explanations; and attempts to reduce AI-assisted cheating.

Disadvantages / Limitations: Depends on an external LLM, which can introduce API cost and computational dependency. It is focused on educational code review rather than a complete multi-language assessment platform.

2. Web-CAT: Automatically Grading Programming Assignments

Author(s): Stephen H. Edwards and Manuel A. Pérez-Quirón (2008)

Algorithm / Methodology: Open-source automated grading system supporting multiple programming languages, customizable assessment strategies, reference tests, automated grading, and manual design assessment.

Advantages: Scalable; extensible; supports multiple languages; provides rapid feedback; encourages students to test their own programs; and combines automated and manual evaluation.

Disadvantages / Limitations: Primarily based on automated grading and testing rather than modern AI-driven understanding of student code. It also does not directly address generative-AI-assisted submissions.

3. Paper-Based Coding Exams and CodeShuffler

Author(s): Hasan Baig and Phillip Bradford (2025)

Algorithm / Methodology: Paper-based coding assessment supported by CodeShuffler, which shuffles correct code lines, numbers them, and generates multiple-choice sequence options.

Advantages: Reduces the direct impact of AI-generated solutions; evaluates code understanding and sequencing; reduces grading workload; and can support large classes.

Disadvantages / Limitations: Does not fully evaluate practical coding activities such as typing, compiling, executing, and debugging. Multiple-choice formats can allow guessing, and students do not receive immediate execution or debugging feedback.

3. Comparative Literature Survey

Paper	Methodology	Advantages	Limitations
GPT-Based Code Review System	GPT-4o, CRM, CCM and RNC with test cases and LLM assessment.	Personalized feedback; AI quality checking; reduced unnecessary reviews.	LLM cost/dependency; focused educational setting.
Web-CAT	Automated grading with customizable tests and assessment strategies.	Scalable, extensible, multi-language support, rapid feedback.	Limited modern AI code understanding; no direct AI-cheating focus.
Paper-Based Coding Exams and CodeShuffler	Shuffled code lines with multiple-choice sequencing.	Tests code understanding; reduces AI answer dependence and grading effort.	Limited practical coding assessment; guessing possible; no execution feedback.

4. Research Gap Identified

- **Integrated assessment:** Existing work often focuses separately on automated grading, AI-based feedback, or AI-resistant assessment. A unified platform combining these functions is needed.
- **Assessment beyond correctness:** A complete platform should consider code quality, logical approach, efficiency, and programming understanding in addition to test-case results.
- **Fair assessment in the AI era:** Generative AI can produce correct code without demonstrating student understanding, creating a need for mechanisms that assess genuine coding ability.
- **Personalized feedback:** AI can improve beginner feedback, but it should be accurate, controlled, and pedagogically useful.
- **Scalability and cost:** LLM-based evaluation can increase computational and API costs, so efficient evaluation pipelines are required.
- **Explainable evaluation:** Students and instructors need understandable reasons for scores, detected errors, and improvement suggestions.

5. Proposed Direction for the AI-Based Coding Assessment Platform

Based on the three papers, the proposed platform can combine automated programming assessment, AI-assisted code analysis, and AI-aware assessment design. The proposed system may include:

- Online coding environment for writing and submitting programs.
- Automated compilation and test-case execution for functional correctness.
- AI-based analysis for logical errors, code quality, and improvement opportunities.
- Personalized learner-friendly feedback without unnecessarily revealing complete answers.
- Assessment mechanisms that evaluate programming understanding in addition to final output.
- Instructor dashboard for question management, student performance, results, and analytics.
- Configurable support for different programming languages and course requirements.

6. References

D.-K. Lee and I. Joe, "A GPT-Based Code Review System With Accurate Feedback for Programming Education," IEEE Access, vol. 13, 2025. DOI: 10.1109/ACCESS.2025.3581139.

S. H. Edwards and M. A. Pérez-Quiñones, "Web-CAT: Automatically Grading Programming Assignments," ITiCSE, 2008.

H. Baig and P. Bradford, "Paper-based coding exams: Overcoming coding assessments and grading challenges in the era of AI Large Language Models like ChatGPT," 2025 ASEE Northeast Section Conference.